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THOX MeshStack Community Mesh: Opt-In User-Contributed Hardware for a Privacy-Centric Shared Computing and Distributed Inference Mesh

The household MeshStack specification pools one owner's devices. This paper extends it to a permissioned community: MeshStack subscribers opt their own hardware in, and the pooled capacity serves models no single household can place, with the coordinator still outside the payload path, transfers still grant-gated, and a blockchain-like ledger confined to the control plane so inference pays no consensus cost.

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ABSTRACT

The household MeshStack specification (Ross and Xaypanya, 2026) pools a single principal's small devices through capacity-priced pipeline sharding, grant-gated parent-chained HMAC KV transfer, and lease-fenced roles, with the coordinator outside the payload path. Its bound is the household: at the published 2/4/6 GB class floors a three-device mesh cannot place a 27B-class model. This paper specifies and evaluates in a virtual lab an opt-in community extension. Authenticated MeshStack subscribers contribute idle hardware under epoch-bearing leases; subscribers whose models exceed household capacity route to the community pool under principal-bound, attenuable grants; and MeshCredits account for contributed memory-time. A blockchain-like infrastructure, an append-only hash-chained ledger with batched Merkle anchoring, records enrollments, grants and credits on the control plane only, so verifiable accounting costs nothing per token. The lab evidence is calculated and simulated, and is labelled as such: under the two-domain admission rule a 27B-class model that is unplaceable on the household trio places in 2 pipeline stages on four 16 GB community nodes, a 70B-class model places in 4 stages on the same pool, all eight privacy drills pass (cross-principal, stale-epoch, fingerprint-mismatch and revocation attempts all fail closed), and ledger appends cost 8.65 microseconds per control-plane entry with zero ledger operations in the inference path. No production community mesh and no community-mesh measurement is claimed. The central privacy risk, prompt recovery from intermediate activations by a malicious stage holder, is stated plainly and bounds what the community pool may be used for until a measured defence exists.

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Executive summary

MeshStack subscribers already pool their own devices. The community extension answers the next question: what happens when a subscriber's model is larger than their household? The answer specified here is a permissioned, opt-in pool of other subscribers' hardware, governed by the same four household invariants, plus a verifiable accounting layer that never touches inference.

2 stages
27B-class model on four 16 GB community nodes; Unplaceable on the household trio at the published floors
CALCULATED

8 / 8
privacy and grant drills pass: cross-principal, attenuation-widening, stale-epoch, fingerprint-mismatch and revocation attempts all rejected
SIMULATED

8.65 µs
per ledger append on the control plane; 0 ledger operations per token or per activation in the inference path
LAB-MEASURED, GENERAL HOST

9.32 µs
parent-chained HMAC keying per 64-token chunk in this lab, corroborating the household measured 9.7 µs
LAB-MEASURED, GENERAL HOST

PRODUCT IMPLICATIONS

First, the community pool is a subscription capability, not a public swarm: enrollment is authenticated, opt-in and revocable, and owner foreground work (P0) preempts contributed work without charge. Second, private cloud-scale models become reachable without cloud inference: a subscriber's request can be served by a pipeline of community stages while plaintext stays on devices the subscriber owns or has explicitly granted, and optional hosted routes remain separate, disclosed and consented. Third, the ledger gives the credit system an audit trail founders and subscribers can verify, without putting a blockchain, or any consensus protocol, in the latency path.

1 Introduction

Household MeshStack demonstrates that a small number of owner-controlled devices can be pooled without treating the home as a data center. The central limitation is scale: the largest model a household can place is bounded by its own aggregate usable memory. The household paper's central negative result makes the bound concrete: a 27B-class model is unplaceable on an Air, Mini and Clip trio at the published class floors. Public cloud inference removes that bound at the cost of data residency, transport confidentiality and execution confidentiality. A community of authenticated MeshStack subscribers and contributors restores scale while retaining the household privacy invariants.

This paper makes four contributions. (1) A research specification for the multi-principal extension: opt-in enrollment under epoch-bearing leases, capacity-

priced two-domain admission scoped across principals, pipeline sharding with whole-request routing as the preferred degenerate case, and grant-gated KV transfer with crypto-shredding. (2) A blockchain-like control-plane design (Section 5) that provides verifiable accounting for grants and MeshCredits with no consensus step and no per-token cost, together with the measured lab overhead of that design. (3) A complete, corrected Virtual Lab: the uploaded v0.1 lab's pipeline function never sharded weights and reported an 8B model that fits whole on one node as needing four nodes; the lab shipped with this paper implements true per-stage sharded admission, a grant-DAG simulator, a churn simulator and the ledger, and every number in Section 7 is read from its `summary.json`. (4) An honest privacy analysis: the design inherits the household mechanisms, and it also inherits, and states, the activation-inversion risk that recent work measures against collaborative inference generally.

No production community mesh is claimed. No community-mesh measurement is claimed. Measured numbers cited from the household paper are labelled MEASURED (household, host class); everything produced by this paper's lab is CALCULATED or SIMULATED, or, for two micro-benchmarks, LAB-MEASURED on a general host and never presented as a device result.

2 Related work

Household MeshStack. Ross and Xaypanya (2026) specify capacity-priced pipeline sharding with two-domain fail-closed admission, a parent-chained HMAC-keyed KV graph gated by a cost model and a live authority path, and lease-fenced roles. Measured on host classes: lossless byte-split zstd compression of real GGUF KV at 1.17×–1.20×, and parent-chained HMAC keying at 9.7 µs per 64-token chunk [1].

Public collaborative inference. Petals demonstrates a public peer-to-peer swarm hosting large models by assigning contiguous layer spans to volunteer servers [2]; its companion systems paper treats fault-tolerant inference over unreliable, join-and-leave consumer devices [3]. The public swarm exposes intermediate activations to peers and is, by its own documentation, unsuitable for confidential data. Prima.cpp is the closest measured household analogue: pipelined-ring parallelism on heterogeneous home devices, with 30B to 70B models over Wi-Fi [4].

The activation-privacy problem. Qu et al. show that a malicious participant in collaborative inference can recover the input prompt from the activation sent by the previous participant, at 88.4% token accuracy in their deepest setting [5]. Noorbakhsh et al. propose information-theoretic privacy adapters that reduce attack success [6]. Any contributed-hardware mesh, including this one, must answer Qu et al. directly; Section 8 does so as a limitation and a usage bound, not as a solved problem.

Community compute and credits. WWW.Serve interconnects LLM services decentrally, with providers choosing when and under what policies they join,

and specifies a credit system in which participants earn by serving and spend by offloading [7]. Lin and Zhang propose an open inference market for contributed hardware and replace proof-of-work's verify asymmetry with earnings-as-collateral against cheating [8]. DICOMPUTE frames distributed inference economics with a full-cost model and a privacy model separating residency, transport confidentiality, execution confidentiality and verifiable accounting; it is a framework with falsifiable hypotheses, not measured results [9]. IslandRun proposes privacy-aware multi-objective orchestration across personal, edge and regional resources and is, by its own discussion, a theoretical framework without empirical evaluation [10].

KV transfer. CacheGen compresses and streams KV over bandwidth-limited links [11]; KIVI quantises KV to 2 bits [12]; KVCodec uses GPU video codecs for remote KV reuse on constrained links [13]. MeshStack's wire position remains the household one: lossless by default at a measured 1.17×–1.20× on real GGUF KV, with lossy tiers a labelled opt-in.

Differentiation. The THOX community mesh is permissioned and account-bound where Petals is public; grant-gated and principal-bound where shared-system designs rely on trust registries; fail-closed on admission, epoch and fingerprint where volunteer systems optimise for availability; and its accounting ledger is a hash-chained control-plane log, not a consensus chain in the serving path.

3 System and threat model

A community mesh is the set of nodes enrolled under authenticated principals who have opted in to contribute or consume capacity. Nodes remain under their owners' administrative control. Links may be LAN, tailnet or wider-area, with the same variable, shared character as the household case. Churn is routine: an owner may drain, move or power off a node at any time, and owner foreground work is P0 and preempts contributed work.

Threats in scope: an honest-but-curious peer inside the community; a malicious stage holder attempting prompt recovery from activations; a departed node returning with a stale lease; a stolen or compromised contributor device; a network observer; a malicious subscriber attempting to reconstruct another principal's state or spend another principal's credits. The coordinator is honest-but-curious for metadata only and is never in the payload path. No node is trusted with another principal's plaintext unless an explicit, live grant authorises it. Fail-closed is the default for admission, grant validation, fingerprint mismatch and ledger divergence.

4 Design

4.1 Enrollment and lease-fenced roles

A contributor authenticates as a MeshStack subscriber, advertises usable capacity (memory, thermal headroom, link class), and receives an epoch-bearing lease granting an InferenceNode or Anchor role in the community pool. The lease is scoped, revocable and recorded in the ledger. Subscribers receive grant-issuance rights limited to their subscription tier. Roles and epochs extend the household secrets graph with community scopes that isolate principals.

4.2 Capacity pricing and two-domain admission

Admission is the household two-domain rule, applied per stage: a stage's share of weights plus its share of attention KV scaled by sequence length must fit the node's usable memory, and the workspace reserve (20% of usable memory in this lab) must remain. Partial admission fails closed. Contributors earn MeshCredits proportional to offered memory-time and completed work (in the lab's calculated example, 352 credits for 16 GB × 10 h plus 32 GB × 6 h at 1 credit per GB-hour); subscribers spend credits or subscription grants. Preempted stage-hours are not charged.

4.3 Pipeline sharding and whole-request routing

Contiguous layer ranges are assigned to stages by a first-fit contiguous planner (Algorithm 1 in the household specification, implemented per-stage in this lab). One activation tensor per token crosses each stage boundary over an authenticated, encrypted tunnel. Whole-request routing to a single node holding the complete model is the preferred degenerate case and is what the planner returns whenever one node admits the whole model.

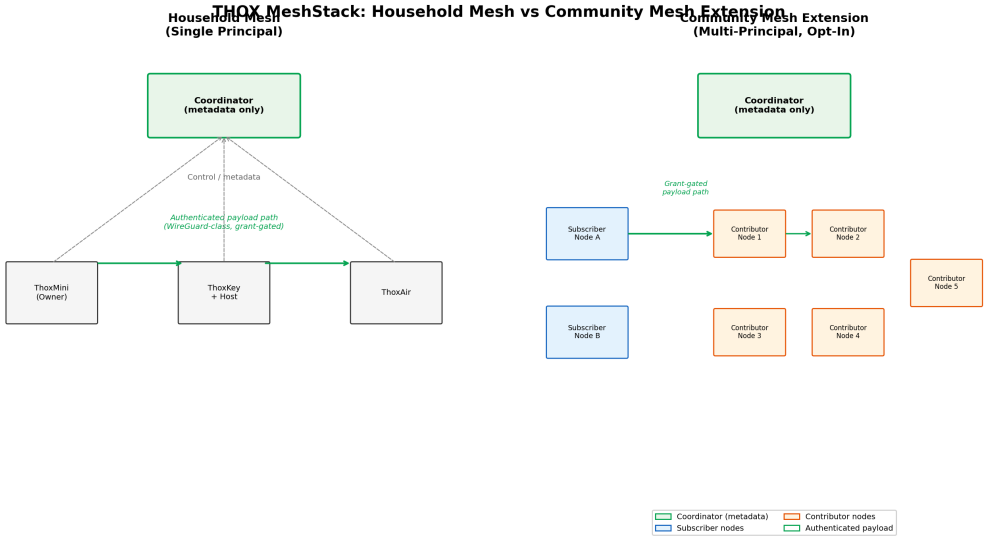


Figure 1. Household mesh (left) versus community mesh extension (right). Solid green arrows are authenticated, grant-gated payload paths; dashed arrows are control and metadata only. The coordinator remains outside the payload path in both cases.

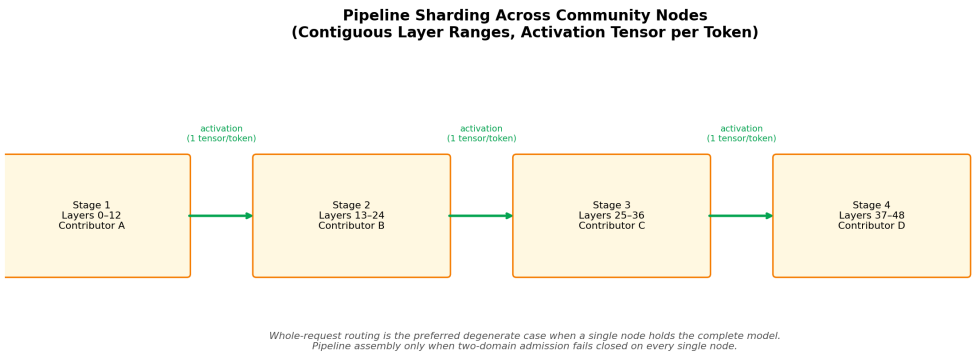


Figure 2. Pipeline sharding places contiguous layer ranges on contributor stages. One activation tensor per token crosses each boundary. Whole-request routing remains the preferred degenerate case.

4.4 Grant-gated KV graph

The KV cache remains a parent-chained, HMAC-keyed graph bound to an authenticated principal and an exact model fingerprint (weights, quantisation, tokenizer, RoPE, layout, engine). Cross-node transfer is gated by the PLACE cost model and by a live authority path in the grant DAG. Grants are attenuable: a derived grant may only narrow its parent's rights, and may never change principal. Consumption-time validation rejects stale epochs. Crypto-shredding deletes a principal's descendants on revocation or lease expiry. Community scopes prevent cross-principal reuse.

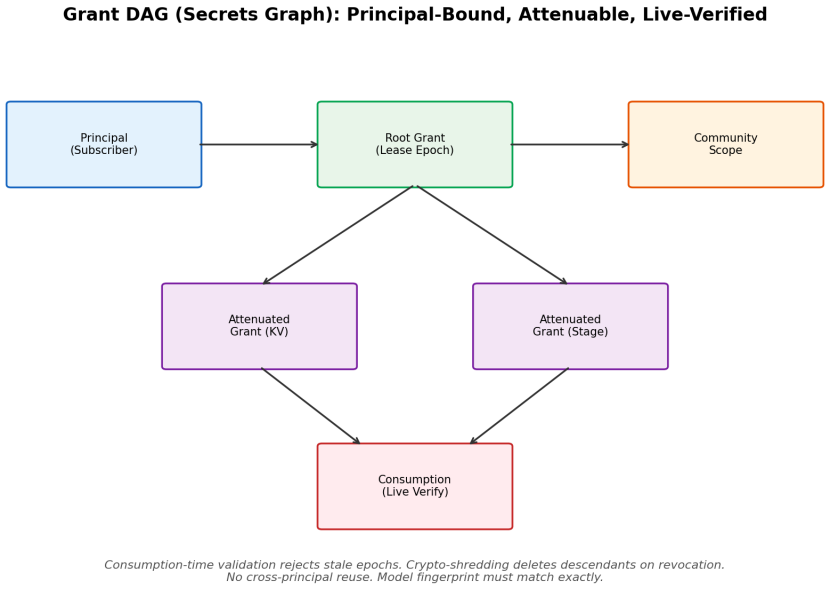


Figure 3. Grant DAG: principal-bound, attenuable grants with consumption-time validation and crypto-shredding on revocation. No cross-principal reuse. The model fingerprint must match exactly.

4.5 Privacy commitments and coordinator placement

Plaintext stays on devices the initiating principal owns or has explicitly granted. The coordinator is not in the payload path; it receives neither model weights nor private inference payloads, only metadata (capacity advertisements, block

identifiers, token counts, load). Peers are authenticated before work is accepted. No cloud inference is required; optional private hosted routes are separate, disclosed and consented, and are how a subscriber reaches a private equivalent of a cloud-scale model when the community pool cannot place it. Data residency, transport confidentiality and execution confidentiality are maintained by principal-bound keying, grant attenuation and encrypted tunnels, subject to the activation-inversion bound stated in Section 8.

5 A blockchain-like control plane without an inference cost

The community mesh needs three properties a blockchain is usually bought for: an append-only history nobody can quietly rewrite, credits that cannot be double-spent or forged, and grant events (issue, attenuate, revoke) that any participant can audit after the fact. It does not need, and must not pay for, distributed consensus in the serving path. A token generated in tens of milliseconds cannot wait on block times, validator rounds or global ordering.

The design therefore takes the blockchain's data structure and leaves its consensus behind. (1) **Hash-chained append-only log.** Every control-plane event (enrollment, lease issue and expiry, grant issue and revocation, credit earn and spend) is an entry chained to the hash of its predecessor, held by the coordinator and mirrored by participants. Rewriting history breaks the chain at the first altered entry. (2) **Batched Merkle anchoring.** Periodically (by count or by time) the coordinator publishes a single Merkle root over a batch of entries. A subscriber verifies their own credit or grant entry with a logarithmic inclusion proof, without downloading the log and without trusting the coordinator's word for that entry. (3) **Local grant validation.** Inference-time grant checks run against a locally held, signed epoch checkpoint, exactly as in Section 4.4. The ledger is evidence about grants, never a step in validating one. (4) **Fail-closed divergence.** If a mirror's chain or a batch root does not verify, the affected scope stops admitting new community work until the divergence is resolved; in-flight owner work is unaffected.

What this deliberately is not: there is no mining, no staking, no token, no smart contract and no cross-community consensus. Ordering inside one community pool is the coordinator's, made tamper-evident by the chain and the anchors rather than tamper-proof by consensus. That is the right trade for a permissioned, account-bound pool whose operator is already identified and accountable, and it is the same trust shape DICOMPUTE's verifiable-accounting leg and the Open Inference Market's earnings-as-collateral mechanism reach for by heavier means [8, 9].

The cost, measured in this lab on a general host, is 8.65 µs per appended entry, a 25.2 ms full verification of a 5,000-entry chain, and 18.44 µs to verify a 13-hash Merkle inclusion proof. A tampered entry in the drill was detected by chain verification. Steady-state inference performs zero ledger operations per token

and per activation: the ledger's latency is paid by enrollment, leasing and accounting events, which are already control-plane operations, and by nothing else.

6 Implementation: the Virtual Lab

The Virtual Lab shipped with this paper (`exp/community_lab.py`, stdlib Python) implements the specification end to end at lab fidelity: the per-stage two-domain planner, the grant DAG with attenuation, principal binding, epoch and fingerprint validation and crypto-shredding, the parent-chained HMAC chunk keying, a churn simulator, the MeshCredits accounting example, and the hash-chained ledger with Merkle anchoring. Five unit tests in `exp/test_lab.py` pass, including the household negative result and the rule that a node running owner foreground work is never used. The lab replaces the uploaded v0.1 draft, whose pipeline function tested the whole model against single nodes and returned a capped count of capable nodes; its capacity table contradicted the draft paper's own table, and neither is carried forward.

The lab is a specification harness, not an inference engine: it computes plans, keys, grants, ledger proofs and accounting, and executes no tokens. Repository review for this paper (2026-11-10, GitHub as ttracx) establishes exactly what exists to build on. The distributed-inference design lives in ttracx/thox-meshstack-inference at dcb81cb5: a self-labelled Python reference package (meshstack_v2: sharding, stage planner, scheduler, kv_graph, secrets_graph) with expanding Rust crates and a llama.cpp-derived stage server; its current HEAD is a stub ship-check loop with live wiring explicitly open, and its multi-user admission tool is, by its own docstring, bounded in-process admission with no routing, auth or engine calls. The paper of record is ttracx/thox-meshstack-paper at b883bb93 (v1.1.0). The production-intent coordinator (identity binding, Stripe subscription state, device registry) is in ttracx/THOX_MeshStack at 6f455990 and binds one user identity; the live virtual device lab is ttracx/thox-virtual-device-lab at 339cc69a (the Thox-ai copy of that name is a stale November 2 duplicate). The sole opt-in sharing precedent in production code is meshstack-desktop at 972711db: a Core AI host model may serve mesh peers only after the owner approves its exact model ID, with the approved list empty by default. Searched and not found in any repository: multi-principal trust or per-principal KV scoping, a community enrollment, consent or revocation service, and any credit or contribution-accounting code; the only money code is Stripe subscription billing. The community extension specified here is therefore presented as what it is: a new design and a new lab, built on the household mechanisms, not a description of shipped community features.

7 Evaluation

Every number below is read from `exp/results/summary.json`, the lab's single evidence source. Sequence length is 4096 throughout; usable memory is the node's stated usable figure; the workspace reserve is 20%.

7.1 Capacity (CALCULATED)

Model	Weights (GB)	Household trio	Single 16 GB	Community 4×16 GB	Community 4×32 GB	Single 64 GB
Qwen3-8B	5.0	3 stages (pipeline)	1 stage (whole-request)	1 stage (whole-request)	1 stage (whole-request)	1 stage (whole-request)
Gemma-3-27B	16.5	Unplaceable	Unplaceable	2 stages (pipeline)	1 stage (whole-request)	1 stage (whole-request)
Qwen3-32B	20.1	Unplaceable	Unplaceable	2 stages (pipeline)	1 stage (whole-request)	1 stage (whole-request)
Llama-3.3-70B	43.2	Unplaceable	Unplaceable	4 stages (pipeline)	2 stages (pipeline)	1 stage (whole-request)
THOX-27B-ref	14.6	Unplaceable	Unplaceable	2 stages (pipeline)	1 stage (whole-request)	1 stage (whole-request)

Table 1. Placement by pool under the two-domain rule. CALCULATED from published model geometry and class floors; no community-mesh measurements are claimed. The household trio uses usable figures of 1.6 / 3.2 / 4.8 GB (80% of the 2 / 4 / 6 GB floors).

Figure 4. Stages required by pool (CALCULATED, seq 4096, 20% reserve)

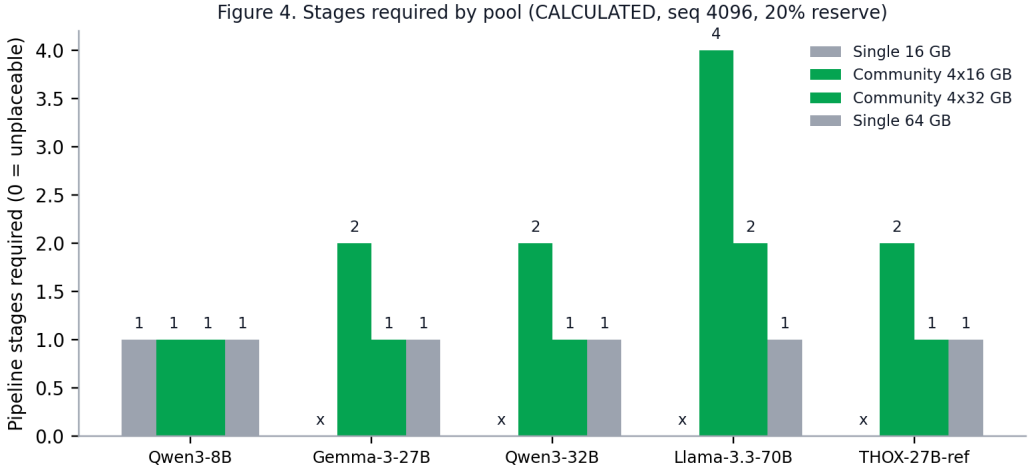


Figure 4. Pipeline stages required by pool. Zero denotes Unplaceable (fail-closed). CALCULATED.

The household negative result reproduces exactly: the 27B reference geometry is Unplaceable on the trio. The community extension removes that bound at modest scale: the same model places in 2 stages on four 16 GB nodes, a 70B-class model places in 4 stages on that pool and in 2 stages on four 32 GB

nodes, and an 8B model that pipelines across the household trio is a single whole-request placement on any community node of 16 GB or more.

7.2 Privacy and grant drills (SIMULATED)

Eight drills run against the grant DAG: a valid KV grant is accepted; attenuation-widening is rejected; a cross-principal grant is rejected; a stale-epoch grant is rejected; a fingerprint-mismatched grant is rejected; an ungranted right is rejected; after root revocation, every descendant is rejected and the shredded set is recorded (kv1, root, stale); and the coordinator's record schema is verified to contain metadata fields only. All eight pass.

7.3 Churn (SIMULATED)

Figure 5. Churn simulation, 30 seeds (SIMULATED)

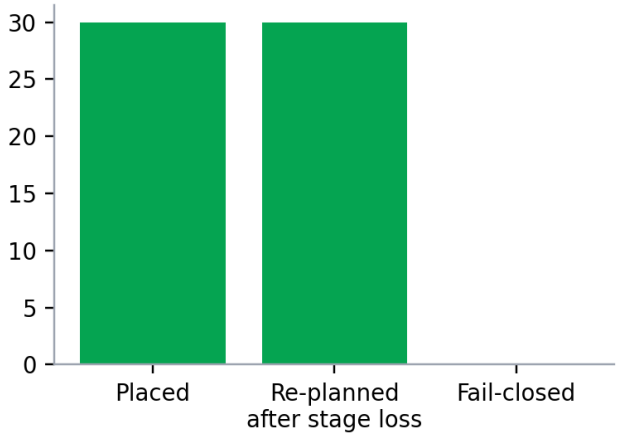


Figure 5. Churn simulation over 30 seeds with mixed 10–50 node pools: every pool placed the 27B-class model, and every pool re-planned after losing its first stage node. SIMULATED.

Fail-closed outcomes occur in the capacity results (the Unplaceable rows of Table 1), which is where this design intends them: at admission, before work starts, rather than mid-generation.

7.4 Keying and ledger overhead (LAB-MEASURED, general host)

Parent-chained HMAC keying costs a median 9.32 µs per 64-token chunk over 2,000 chunks in this lab, corroborating the household measured 9.7 µs on its host class. The ledger costs are reported in Section 5. A synthetic-tensor compression drill in the lab round-trips bit-exactly; the paper's wire ratio remains the household MEASURED 1.17×–1.20× on real GGUF KV, and no lab synthetic ratio is presented as a wire result.

8 Discussion and limitations

Activation inversion is the binding limitation. A stage holder sees the activations it computes on. Qu et al. measure prompt recovery from exactly those activations [5]. Encrypted tunnels protect activations from observers, not from the stage that must compute on them. Until a defence in the class of

Noorbakhsh et al. [6] is implemented and measured in this system, the community pool is specified for workloads whose prompts the subscriber accepts exposing, in activation form, to authenticated fellow subscribers, and confidential workloads remain household-only. The paper does not soften this: it is the honest price of pipeline inference on other people's hardware, and Petals' own documentation reaches the same conclusion for its public swarm.

Further limitations. Engine evidence is thin and inherited from the household work. WAN latency and bandwidth across a real community are unmeasured; the churn simulation assumes re-planning is cheap relative to generation. Free-riding, credit-market gaming and Sybil enrollment are mitigated by account binding and earnings held against audit, not solved. There is no attested execution and no differential privacy on activations. Side channels are unexamined. Control-plane ownership under wide-area churn is undecided, and the ledger's single-operator ordering is accountable rather than trustless by design. KV remains a cache, not semantic memory.

The negative capacity result of the household paper remains the strongest argument for the extension: when household floors cannot place a useful model, a permissioned community pool is the remaining local-first path, and this paper specifies that path with its costs and its privacy bound stated.

9 Conclusion

The household MeshStack specification pools devices under one principal. The community extension pools devices under many authenticated principals while preserving the same privacy invariants, fail-closed admission, grant-gated transfers and lease fencing, and adds a blockchain-like accounting ledger that lives entirely on the control plane. MeshStack subscribers can therefore reach models larger than any single household can hold, on hardware fellow subscribers opted in, without placing the coordinator in the payload path, without traditional cloud inference, and without paying a consensus cost in the inference path. Implementation against real community hardware, and measurement of the privacy defence the design requires, remain future work and are stated as such.

Appendix A. Reproducibility

Run `python3 exp/community_lab.py` to regenerate `exp/results/summary.json`; run `python3 exp/test_lab.py` for the five unit tests; figures are regenerated by `wp/figures.py` from the same summary. Measured numbers cited from the household paper are taken from that report (DOI 10.5281/zenodo.23969070, v1.1.0) and labelled with their host class. The TikZ sources for Figures 1–3 ship in the artifact bundle. No production community mesh, device measurement or community-mesh measurement is claimed anywhere in this paper.

Appendix B. Authorship and disclosure

Authors: Craig Ross, Chief Executive Officer, THOX.ai LLC, and Tommy Xaypanya, Chief Technology Officer and Co-Founder, THOX.ai LLC (corresponding author; ORCID 0009-0000-4425-9565). Synthetic agents assisted with literature search, lab implementation, outline synthesis and drafting under the direction of the human authors. All technical claims, evidence labels and design decisions were reviewed and remain the responsibility of the human authors.

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